

Compound IF Statements – Practical Guide (Solutions)

Relational operators:

Relational Operator Symbol	Description	Example
=	Equal to	numDays = 5 day = "Monday"
<	Less than	studMarks < 50
>	Greater than	studMarks > 75
<=	Less than or equal to	age <= 18
>=	Greater than or equal to	age >= 60
<>	Not equal to	numMinutes <> 60 cityName <> "Gqeberha"

Structure of a Simple IF Statement:

```
IF variable relational operator value THEN  
    Calculation, assigning or display instruction  
ENDIF
```

Example of a Simple IF Statement:

```
IF age >= 18 THEN  
    DISPLAY "You can vote"  
ENDIF
```

Structure of a Compound IF Statement:

```
IF variable relational operator value logical operator variable relational operator value THEN  
    Calculation, assigning or display instruction  
ENDIF
```

Example of a Compound IF Statement:

```
IF age >= 60 and citizenship = "South African" THEN  
    yearGrant = 2000 * 12  
ENDIF
```

Steps on how to write an IF Statement:

1. Identify and write the variable that is checked in the sentence or table column.
2. Identify and write the relational operator used in the sentence or table row.
3. Identify and write the value that is checked with the variable.
4. Identify and write the statement required inside the IF Statement between THEN and ENDIF.

****Note:** If the IF Statement is a compound of two conditions, then repeat step 1, 2 and 3.

Read the scenario below and write the pseudocode IF Statements for the program requirements:

The Rising Skills Center runs short courses and offers small incentives to encourage attendance and achievement. The Center needs a program to collect each applicant's details and calculate the total fees, allowances, and total bonuses based on the applicant's details.

The program must accept the applicant's name, age, highest grade passed (1 to 12), number of course days attended, physical state, and distance from the Center in kilometers.

The following pseudocode has already been written:

BEGIN

 DECLARE name, status AS string

 DECLARE age, numDays, highestGrade AS integer

 DECLARE distance, totalFee, allowance, totalBonus AS real

 DISPLAY "Enter the applicant's name"

 ACCEPT name

 DISPLAY "Enter the applicant's age"

 ACCEPT age

 DISPLAY "Enter the applicant's highest grade passed"

 ACCEPT highestGrade

 DISPLAY "Enter the number of course days attended by the applicant"

 ACCEPT numDays

 DISPLAY "Enter the applicant's disability status"

 ACCEPT status

 DISPLAY "Enter the applicant's distance from the Center"

 ACCEPT distance

Program Requirements

Simple IF Statements:

1. If the applicant is under 16 years old, show the message: "Too Young" together with the applicant's age.

Solution:

```
IF age < 16 THEN
    DISPLAY "Too Young", age
ENDIF
```

2. If the applicant has completed grade 12, display the applicant's name with the message "Certificate Bonus Awarded".

Solution:

```
IF highestGrade = 12 THEN
    DISPLAY "Certificate Bonus Awarded", name
ENDIF
```

3. Calculate the Total Fee based on the following:

DISABILITY STATUS	DAILY FEE PER COURSE DAY
Disabled	R150
Not Disabled	R300
Chronic Disease	R200

Solution:

```
IF status = "Disabled" THEN
    totalFee = 150 * numDays
ENDIF
```

```
IF status = "Not Disabled" THEN
    totalFee = 300 * numDays
ENDIF
```

```
IF status = "Chronic Disease" THEN
    totalFee = 200 * numDays
ENDIF
```

4. Calculate the Total Bonus based on the following:

AGE	DAILY BONUS PER COURSE DAY
More than 65	R30
Less than or equal to 20	R15

Solution:

```
IF age > 65 THEN
    totalBonus = 30 * numDays
ENDIF
```

```
IF age <= 20 THEN
    totalBonus = 15 * numDays
ENDIF
```

Compound IF Statements:

5. If the applicant has passed at least grade 8 and has attended at least 20 course days, calculate the allowance by multiplying the number of days attended by 2.

Solution:

```
IF highestGrade >= 8 and numDays >= 20 THEN
    allowance = numDays * 2
ENDIF
```

6. If the applicant is older than 21 and lives more than 30 km away in distance, calculate the total bonus by adding R300 to the allowance.

Solution:

```
IF age > 21 and distance > 30 THEN
    totalBonus = allowance + 300
ENDIF
```

7. If the applicant has passed grade 5 and has attended at least 15 days, output the applicant's grade with the message "Reduced Fee Applied".

Solution:

```
IF highestGrade >= 5 and numDays >= 15 THEN
    DISPLAY "Reduced Fee Applied", highestGrade
ENDIF
```

8. If the applicant travels between 5 km and 20 km in distance to attend classes, show the distance traveled.

Solution:

```
IF distance >= 5 and distance <= 20 THEN
    DISPLAY "The distance is ", distance
ENDIF
```

9. If the number of course days attended is from 10 to 15, display the total bonus.

Solution:

```
IF numDays >= 10 and numDays <= 15 THEN
    DISPLAY "The total bonus is ", totalBonus
ENDIF
```

10. If the applicant lives within 10 km distance and has attended 25 days or more, the allowance calculation should be 15% of the total fee.

Solution:

```
IF distance <= 10 and numDays >= 25 THEN
    allowance = totalFee * 0.15
ENDIF
```

11. Display a message based on the distance:

DISTANCE IN KM	MESSAGE
From 3 km to 10 km	Close to the Center
More than 40 km but less than 60 km	Far from the Center
> 75 km	Not within the Center's region

Solution:

```
IF distance >= 3 and distance <= 10 THEN
    DISPLAY "Close to the Center"
ENDIF
```

```
IF distance > 40 and distance < 60 THEN
    DISPLAY "Far from the Center"
ENDIF
```

```
IF distance > 75 THEN
    DISPLAY "Not within the Center's region"
ENDIF
```

12. If the applicant is older than 21, and has passed at least grade 10, output the applicant's name with the message "Advanced Discount Granted".

Solution:

```
IF age > 21 and highestGrade >= 10 THEN
    DISPLAY "Advanced Discount Granted", name
ENDIF
```